MAHATMA GANDHI UNIVERSITY
THIRD SEMESTER UG PRACTICAL EXAMINATION
Course:MG3DSCCHE 202- Inorganic and Organic Chemistry Practical

Model Questions

Time **-3 hr.** Total marks: **35**

Part-I (10 marks)

Conduct **one** Experiment with √ mark (Select One)

1. Estimate the mass of Fe^{2+} in the whole of the given solution of ferrous sulphate. You are supplied with approximately 0.02M KMnO4 and A R Mohr s salt crystal.
2. Estimate the mass of Oxalic acid in the whole of the given solution of oxalic acid solution. You are provided with 0.02 M KMnO4 and crystals of analytical grade oxalic acid.
3. Estimate the mass of Fe^{2+} in the whole of the given solution of ferrous sulfate (whose molarity will be 0.1M after making up to 100ml). You are provided with AR potassium dichromate crystals (using internal indicator)
4. Estimate the mass of Fe^{2+} in the whole of the given solution of ferrous sulfate (whose molarity will be 0.1M after making up to 100ml) using external indicator. You are provided with AR potassium dichromate crystals.

Part-II (5 marks)

Write the theory and procedure for the organic synthesis (Select One)

5. Preparation of m-Dinitrobenzene from Nitrobenzene

OR

6. Preparation of Phenyl Benzoate from Phenol

Part-III (5 marks)

Write the principle and Procedure for the given estimation (Select One)

7. To Estimate the mass of Oxalic acid in the whole of the given solution of oxalic acid solution. You are provided with 0.02 M KMnO4 and crystals of analytical grade oxalic acid.

8. To Estimate the mass of Fe^{2+} in the whole of the given solution of ferrous sulfate (whose molarity will be 0.1M after making up to 100ml). You are provided with AR potassium dichromate crystals (using internal indicator)

11. Viva voce **(5 marks)**
12. Lab report **(10 marks)**

Instructions

1. Give one experiment from part-I.

2. A minimum of 3 experiments from Part I (Titration) and 2 from Part II (organic synthesis) along with determination of physical constants are needed in the record.